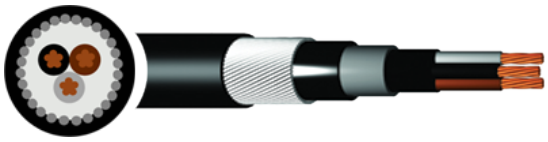
# BS6724 Mains & Control Cable 1kV - XLPE, SWA, LSZH - 1.5mm² to 16mm²



## Description

Used in power networks, indoor, outdoor, underground and in cable ducting. For installation where fire, smoke emissions and toxic fumes create a potential threat to life and equipment.

They are ideal for public buildings like airports and railway stations, computer rooms, datacentres, where escape is limited and fire safety is critical.

## Key Features

**Voltage Rating**
600/1000 Volts

**Minimum Bending Radius**
Fixed: 6 x overall diameter

**Flame Retardancy**
BS EN/IEC 60332-1-2
BS EN/IEC 60332-3-24

**Temperature Limits**
Maximum operating temperature: +90°C

## Core Colours

2 core - Brown, Blue

3 core - Brown, Black, Grey

3 core (up to 70mm²) - Brown, Blue, Green Yellow

4 core - Brown, Black, Grey, Blue

5 core and above (up to 6mm²) - White with Black numbers

2, 3, 4 or 5 core (1.5mm² & 2.5mm²) - White

Also available as

Brown, Black, Grey, Blue, Green Yellow

## Standards

- BS EN/IEC 60332-3-24 (cat C)
- BS6724
- BS EN/IEC 60332-1-2
- BS EN/IEC 61034-2
- Acid Gas Emission to BS EN/IEC 60754-1/2
- BS EN/IEC 60228

## Construction

- **Conductor**: Class 2 stranded copper conductor
- **Insulation**: Cross Linked polyethylene (XLPE)
- **Bedding**: Low Smoke Zero Halogen (LSZH)
- **Armour**: Steel Wire Armour (SWA)
- **Outer Sheath**: Low Smoke Zero Halogen (LSZH)
- **Sheath Colour**: Black

## Training & Testing Lab

AN ISO/IEC 17025 ACCREDITED FACILITY

**Cleveland Cable Test & Training Lab**
Our state-of-the-art cable testing facility ensures that every cable meets the highest standards of quality and compliance through continuous, rigorous testing.

ISO/IEC 17025 Accredited Testing Laboratory
UKAS TESTING 31196

**CPR**
Cleveland Cable Company is committed to compliance with the Construction Products Regulation (CPR). Where applicable, all cables manufactured after 1st July 2017 have been assessed in accordance with CPR requirements, with full supporting documentation available.

## Our Sustainability Commitment

We are committed to the journey to Net Zero as a business partner, an employer and a community member.

By thinking and acting sustainably, we deliver excellent customer service while reducing carbon emissions in collaboration with our customers and suppliers.

CLEVELAND CABLE COMPANY

**ecovadis**
Cleveland Cable Company has been independently assessed by EcoVadis, a globally recognised provider of business sustainability ratings. Our score places us among the top 35% of companies evaluated worldwide, reflecting our strong commitment to environmental, social, and ethical performance

ecovadis

## BS6724 Mains & Control Cable 1kV - XLPE, SWA, LSZH - 1.5mm² to 16mm² - Dimensions

| Reference | Conductor Size (mm2) | No Of Cores | Stranding(mm) | Overall Diameter(mm) | Weight(Kg/Km) | Nylon Cleat Size | Gland Size |
|---|---|---|---|---|---|---|---|
| LSF2X1/5 | 1.5 | 2 | 7/0.53 | 11 | 260 | 0.5 | 20/16 |
| LSF3X1/5 | 1.5 | 3 | 7/0.53 | 11.2 | 295 | 0.5 | 20/16 |
| LSF2EX1/5 | 1.5 | 3 | 7/0.53 | 11.2 | 268 | 0.5 | 20/16 |
| LSF4X1/5 | 1.5 | 4 | 7/0.53 | 12.5 | 350 | 0.5 | 20S |
| LSF5X1/5 | 1.5 | 5 | 7/0.53 | 12.9 | 362 | 0.6 | 20S |
| LSF7X1/5 | 1.5 | 7 | 7/0.53 | 13.7 | 398 | 0.6 | 20S |
| LSF10X1/5 | 1.5 | 10 | 7/0.53 | 18 | 650 | 0.8 | 20 |
| LSF12X1/5 | 1.5 | 12 | 7/0.53 | 18 | 680 | 0.8 | 20 |
| LSF19X1/5 | 1.5 | 19 | 7/0.53 | 20.6 | 885 | 0.9 | 25 |
| LSF27X1/5 | 1.5 | 27 | 7/0.53 | 25.1 | 1310 | 1 | 32 |
| LSF37X1/5 | 1.5 | 37 | 7/0.53 | 27.5 | 1590 | 1.1 | 32 |
| LSF48X1/5 | 1.5 | 48 | 7/0.53 | 31 | 1958 | 1.4 | 32 |
| LSF2X2/5 | 2.5 | 2 | 7/0.67 | 12.2 | 270 | 0.5 | 20S |
| LSF3X2/5 | 2.5 | 3 | 7/0.67 | 12.8 | 360 | 0.5 | 20S |
| LSF2EX2/5 | 2.5 | 3 | 7/0.67 | 12.8 | 340 | 0.5 | 20S |
| LSF4X2/5 | 2.5 | 4 | 7/0.67 | 13.5 | 410 | 0.6 | 20S |
| LSF5X2/5 | 2.5 | 5 | 7/0.67 | 14.7 | 435 | 0.6 | 20 |
| LSF7X2/5 | 2.5 | 7 | 7/0.67 | 15.6 | 520 | 0.7 | 20 |
| LSF10X2/5 | 2.5 | 10 | 7/0.67 | 20 | 250 | 0.8 | 25 |
| LSF12X2/5 | 2.5 | 12 | 7/0.67 | 21 | 905 | 0.9 | 25 |
| LSF19X2/5 | 2.5 | 19 | 7/0.67 | 25 | 1360 | 1 | 32 |
| LSF27X2/5 | 2.5 | 27 | 7/0.67 | 30 | 1760 | 1.2 | 32 |
| LSF37X2/5 | 2.5 | 37 | 7/0.67 | 33 | 2185 | 1.4 | 40 |
| LSF48X2/5 | 2.5 | 48 | 7/0.67 | 37.3 | 3003 | 1.6 | 40 |
| LSF2X4 | 4 | 2 | 7/0.85 | 13.1 | 381 | 0.6 | 20S |
| LSF3X4 | 4 | 3 | 7/0.85 | 13.7 | 435 | 0.6 | 20S |
| LSF2EX4 | 4 | 3 | 7/0.85 | 13.7 | 460 | 0.6 | 20S |
| LSF4X4 | 4 | 4 | 7/0.85 | 14.8 | 495 | 0.6 | 20 |
| LSF5X4 | 4 | 5 | 7/0.85 | 16.4 | 583 | 0.7 | 20 |
| LSF7X4 | 4 | 7 | 7/0.85 | 18.2 | 760 | 0.8 | 20 |
| LSF12X4 | 4 | 12 | 7/0.85 | 24.2 | 1266 | 1 | 25 |
| LSF19X4 | 4 | 19 | 7/0.85 | 27.6 | 1701 | 1.2 | 32 |
| LSF27X4 | 4 | 27 | 7/0.85 | 32.3 | 2347 | 1.4 | 40 |
| LSF2X6 | 6 | 2 | 7/1.04 | 14.1 | 405 | 0.6 | 20 |
| LSF3X6 | 6 | 3 | 7/1.04 | 14.8 | 490 | 0.6 | 20 |
| LSF2EX6 | 6 | 3 | 7/1.04 | 14.8 | 639 | 0.6 | 20 |
| LSF4X6 | 6 | 4 | 7/1.04 | 17.3 | 670 | 0.7 | 20 |

| Reference | Conductor Size (mm2) | No Of Cores | Stranding(mm) | Overall Diameter(mm) | Weight(Kg/Km) | Nylon Cleat Size | Gland Size |
|---|---|---|---|---|---|---|---|
| LSF5X6 | 6 | 5 | 7/1.04 | 18.4 | 823 | 0.8 | 20 |
| LSF7X6 | 6 | 7 | 7/1.04 | 21.9 | 1100 | 0.9 | 25 |
| LSF2X10 | 10 | 2 | 7/1.35 | 16.1 | 600 | 0.7 | 20 |
| LSF | 10 | 3 | 7/1.35 | 17.9 | 750 | 0.8 | 20 |
| LSF4X10 | 10 | 4 | 7/1.35 | 19.3 | 885 | 0.8 | 25 |
| LSF5X10 | 10 | 5 | 7/1.35 | 20.9 | 1106 | 0.9 | 25 |
| LSF7X10 | 10 | 7 | 7/1.35 | 25 | 1500 | 1 | 25 |
| LSF2X16 | 16 | 2 | 7/1.70 | 19 | 905 | 0.8 | 25 |
| LSF | 16 | 3 | 7/1.70 | 20 | 990 | 0.8 | 25 |
| LSF4X16 | 16 | 4 | 7/1.70 | 22 | 1195 | 0.9 | 25 |
| LSF5X16 | 16 | 5 | 7/1.70 | 25.2 | 1695 | 1 | 25 |
| LSF7X16 | 16 | 7 | 7/1.70 | 28.1 | 2150 | 1.2 | 32 |

**Multicore armoured 90 °C thermosetting insulated cables. Reproduced from *BS7671:2018 Wiring Regulations***

## TABLE 4E4A

### CURRENT-CARRYING CAPACITY (amps)

Ambient temperature: 30°C
Ground ambient temperature: 20°C
Conductor operating temperature: 90°C

| Conductor cross- sectional area | Reference Method C (clipped direct) | | Reference Method E (in free air or on a perforated cable tray etc, horizontal or vertical) | | Reference Method D (direct in ground or in ducting in ground, in or around buildings) | |
|---|---|---|---|---|---|---|
| | 1 two-core cable single-phase AC or DC | 1 three- or 1 four- core cable, three- phase AC | 1 two-core cable single-phase AC or DC | 1 three- or 1 four- core cable, three- phase AC | 1 two-core cable single-phase AC or DC | 1 three- or 1 four- core cable, three- phase AC |
| mm2 | (A) | (A) | (A) | (A) | (A) | (A) |
| 1.5 | 27 | 23 | 29 | 25 | 25 | 21 |
| 2.5 | 36 | 31 | 39 | 33 | 33 | 28 |
| 4 | 49 | 42 | 52 | 44 | 43 | 36 |
| 6 | 62 | 53 | 66 | 56 | 53 | 44 |
| 10 | 85 | 73 | 90 | 78 | 71 | 58 |
| 16 | 110 | 94 | 115 | 99 | 91 | 75 |
| 25 | 146 | 124 | 152 | 131 | 116 | 96 |
| 35 | 180 | 154 | 188 | 162 | 139 | 115 |
| 50 | 219 | 187 | 228 | 197 | 164 | 135 |
| 70 | 279 | 238 | 291 | 251 | 203 | 167 |
| 95 | 338 | 289 | 354 | 304 | 239 | 197 |
| 120 | 392 | 335 | 410 | 353 | 271 | 223 |
| 150 | 451 | 386 | 472 | 406 | 306 | 251 |
| 185 | 515 | 441 | 539 | 463 | 343 | 281 |
| 240 | 607 | 520 | 636 | 546 | 395 | 324 |
| 300 | 698 | 599 | 732 | 628 | 446 | 365 |
| 400 | 787 | 673 | 847 | 728 | | |

*1.Where it is intended to connect the cables in this table to equipment or accessories designed to operate at a temperature lower than the maximum operating temperature of the cable, the cables should be rated at the maximum operating temperature of the equipment or accessory (see Regulation 512.1.5).*
*2.Where it is intended to group a cable in this tablewith other cables, the cable should be rated at the lowest of the maximum operating temperatures of any of the cables in the group (see Regulation 512.1.5).*

## TABLE 4E4B

### VOLTAGE DROP (per ampere per metre)

Conductor operating temperature:90°C

| Conductor cross sectional area | Two-core cable DC | Two-core cable, single-phase AC | | | Three- or four-core cable, three-phase AC | | |
|---|---|---|---|---|---|---|---|
| (mm2) | (mV/Nm) | (mV/Nm) | | | (mV/Nm) | | |
| 1.5 | 31 | 31 | | | 27 | | |
| 2.5 | 19 | 19 | | | 16 | | |
| 4 | 12 | 12 | | | 10 | | |
| 6 | 7.9 | 7.9 | | | 6.8 | | |
| 10 | 4.7 | 4.7 | | | 4.0 | | |
| 16 | 2.9 | 2.9 | | | 2.5 | | |
| | | R | X | Z | R | X | Z |
| 25 | 1.85 | 1.85 | 0.160 | 1.90 | 1.60 | 0.140 | 1.65 |
| 35 | 1.35 | 1.35 | 0.155 | 1.35 | 1.15 | 0.135 | 1.15 |
| 50 | 0.98 | 0.99 | 0.155 | 1.00 | 0.86 | 0.135 | 0.87 |
| 70 | 0.67 | 0.67 | 0.150 | 0.69 | 0.59 | 0.130 | 0.60 |
| 95 | 0.49 | 0.50 | 0.150 | 0.52 | 0.43 | 0.130 | 0.45 |
| 120 | 0.39 | 0.40 | 0.145 | 0.42 | 0.34 | 0.130 | 0.37 |
| 150 | 0.31 | 0.32 | 0.145 | 0.35 | 0.28 | 0.125 | 0.30 |
| 185 | 0.25 | 0.26 | 0.145 | 0.29 | 0.22 | 0.125 | 0.26 |
| 240 | 0.195 | 0.20 | 0.140 | 0.24 | 0.175 | 0.125 | 0.21 |
| 300 | 0.155 | 0.16 | 0.140 | 0.21 | 0.140 | 0.120 | 0.185 |
| 400 | 0.120 | 0.13 | 0.140 | 0.190 | 0.115 | 0.120 | 0.165 |

THE INFORMATION CONTAINED WITHIN THIS DATASHEET IS FOR GUIDANCE ONLY AND IS SUBJECT TO CHANGE WITHOUT NOTICE OR LIABILITY. WE BELIEVE THE INFORMATION IS CORRECT AT THE TIME OF PUBLICATION. PLEASE NOTE WHEN SELECTING CABLE ACCESSORIES THAT ACTUAL CABLE DIMENSIONS MAY VARY DUE TO MANUFACTURING TOLERANCES.